Ashwin Chembu

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PROFILE

Full-stack product engineer connecting React and TypeScript interfaces with Python and Node.js services, authentication, and cloud data. Builds complete workflows with role permissions and layered validation.

TECHNICAL SKILLS

Languages / Frontend: TypeScript, JavaScript, Python, React, Next.js, Tailwind CSS, Capacitor
Backend / Data: Node.js, Express, FastAPI, REST APIs, OAuth, WebSockets, MongoDB, Firebase, AWS S3
Testing / Delivery: Vitest, React Testing Library, Firebase emulators, Playwright, Git, Vite

EXPERIENCE

Business Technology Solutions Associate | ZS Associates — **South San Francisco, CA | Jun 2024-Present**

- Built **React**/FastAPI RAG assistants with AWS document ingestion and Redis-backed services, connecting uploaded knowledge to searchable enterprise workflows so teams could retrieve context through usable product interfaces.
- Built Python/Flask endpoints connecting AWS S3 and Smartsheet APIs, coordinating document storage and workflow handoffs so teams could ingest, organize, and evaluate files through integrated cloud services.
- Implemented **React**/JavaScript account-plan submission and manager approval with CRM-backed data services, timestamps, review history, and confirmation paths, simplifying field-user actions while preserving review traceability.
- Added cached-section recovery and explicit retries to **React** CRM views, preserving last-good data after refresh failures and ignoring obsolete requests; **five** regression tests covered initial errors, retries, and selected-plan changes.
- Built a **Node.js** provider-lookup workbench with exact identifier search, Jaro-Winkler address ranking, bulk uploads, and session-bound exports; validated authentication, isolation, input checks, and export safety with **10** local tests.

Founder and Lead Developer | Tallyrus — **Jun 2023-Sep 2025**

- Designed **React** rubric and essay-review interfaces that connect teacher materials to generated feedback, incorporating educator testing and iteration so users could inspect writing quality while retaining consistent scoring logic.
- Integrated **React**/**TypeScript** and Express product workflows with Google OAuth, protected routes, MongoDB persistence, and secure sessions, connecting authenticated educator access to document and essay-feedback operations.
- Implemented a Flask grading API that validates rubric, essay, prompt, and historical-example inputs before generating JSON feedback, connecting model requests to the review workflow without treating automated output as a final grade.
- Connected Express/Multer upload routes for teacher files, rubrics, and historical essays to AWS S3 storage and MongoDB assignment records, making educator materials available to downstream grading and review workflows.
- Integrated LangChain/OpenAI embeddings with NumPy normalization and FAISS similarity search, combining retrieved historical essays with teacher rubrics and prompts in GPT-4o requests to return contextual feedback for educator review.

Data Analyst / Web Development Lead | Tech4Good at UCSC — **Mar 2023-Jun 2025**

- Built **TypeScript**/Figma API chat, text-entry, polling, and Excel/XLSX export workflows in a collaborator-owned FigJam repository, enabling research and classroom teams to collect, discuss, and export structured participant responses.
- Implemented **20+** attributed Node/Express REST routes with MongoDB persistence and logging for chat, text-entry, and polling workflows, supporting threaded messages, anonymous responses, editing, voting, and admin controls.
- Separated chat, text-entry, and polling state across **three MongoDB databases**, using change streams, populated message logs, and persisted voter records to synchronize threads, edits, group responses, and reveal/reset operations.
- Built a responsive Angular career-exploration component for top-paying industries, implementing distinct desktop, tablet, and phone layouts so students could inspect industry information across screen sizes.

PROJECTS

Sylk | Multi-Tenant Supply-Chain Platform — **Private organization product | 2025-Present**

- Built **React** 19/**TypeScript** supply-chain workflows with **Firebase** for organization onboarding, line-item shipments, fulfillment, files, and real-time chat, connecting retailer and manufacturer roles through a shared operational product.
- Enforced organization isolation and role permissions through Firestore and Storage rules, separating retailer-only financial data from shared manufacturing workflows and using Cloud Functions to validate fulfillment transitions.
- Packaged the full product as a Capacitor Android app, adding native-safe routing, **Firebase** Authentication, safe-area handling, mobile file picking, and phone-specific navigation across orders, inventory, billing, chat, and team access.

Lume | AI Meeting Intelligence — **Dec 2025-Jan 2026**

- Built a **React** and Node/Express meeting-intelligence platform with MongoDB records and AWS S3 file storage, connecting meeting capture, uploaded documents, transcript-aware search, and project questions in one reusable workflow.
- Designed a RAG engine with vector retrieval, intent routing, evidence extraction, and generator/critic refinement, combining relevant meeting and file context into citation-backed answers that users can trace to supporting sources.
- Connected meeting capture, document search, and reusable project context through responsive **React** interfaces and WebSocket interactions, allowing transcript and file workflows to share the same full-stack meeting-intelligence product.

3D Commerce Experience | Interactive Storefront — **Public UI prototype**

- Built a **React**/**TypeScript** storefront with reusable product cards, navigation, modals, and responsive layouts, connecting product discovery, search, cart, checkout, and account pages into a complete interactive commerce prototype.
- Developed a Three.js product configurator with GLB assets and Draco decoding, adding independent garment customization, color, and placement controls so shoppers could inspect and customize a varsity jacket from multiple viewing angles.
- Improved mobile WebGL recovery by preloading poster, GLB, and Draco assets, delaying the canvas until visible pixels rendered, and providing retryable poster fallback after context loss; validated desktop/mobile views and production builds.

EDUCATION

University of California, Davis | B.S. Computer Science | GPA: 3.7 — **Davis, CA | Graduated Jun 2025**

LINKS

Portfolio: https://ashchembu.com | LinkedIn: linkedin.com/in/ashwin-chembu-a5292b205 | GitHub: github.com/ashwinchembu